

futures__emsx__strategy

API & Component Contracts

Reference documentation

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1 Overview

1.1 Architectural contract

The system is partitioned so each layer depends only on the layer above it on the event flow, never the other way:

```
Bloomberg market data
--> Market Data Service      (ticks, bars)
--> Strategy                  (TargetPosition)
--> Order Manager            (OrderIntent)
--> Pre-Trade Risk Gateway   (RiskDecision)
--> Execution Adapter        (ExecutionAck)
--> EMSX                     (route updates, fills)
--> Portfolio & Reconciler   (PositionBook, FillLedger, PnL)
```

Hard rules:

- Strategies emit only **TargetPosition**. They never construct orders, never call EMSX, never know about routes.
- Order intents pass through **PreTradeRiskGateway** before reaching any **ExecutionAdapter**.
- Execution adapters are the only modules that import **blpapi**. Other layers work without it.
- Every **OrderIntent** has a deterministic **idempotency_key** (SHA-1 of strategy + instrument + bar-end + side + qty), so replays and restarts cannot duplicate orders.

1.2 Event types (canonical)

Event	Producer	Consumer(s)
MarketTick	Market data service	Bar builder, tick store, stale monitor
BarClosed	Bar builder	Strategy
TargetPosition	Strategy	Order manager
OrderIntent	Order manager	Risk gateway
RiskDecision	Risk gateway	Execution service
ExecutionAck	Execution adapter (sync)	Order lifecycle
ExecutionUpdate	Execution adapter (async subs)	Lifecycle, working book, monitoring
FillUpdate	Execution adapter (async subs)	Position book, PnL, fill ledger
KillSwitchEvent	Kill switch	All services (refuse to trade)

1.3 Notation used in this document

Each entry contains:

- Purpose** one-line summary.
- Signature** full Python signature.
- Pre** caller-side preconditions.
- Post** observable effects after a successful call.
- Raises** exceptions raised.

When an item has no preconditions worth stating, the **Pre** field is omitted.

2 Module core/

Foundational primitives: clocks, canonical events, enums, exceptions, identifiers, logging.

2.1 core.clock

Clock (abstract)

Purpose	Time source. Decouples wall-clock from simulation/replay.
Signature	<code>class Clock(ABC): def now(self) -> datetime</code>
Post	Returns timezone-aware UTC <code>datetime</code> .

SystemClock

Purpose	Wall-clock implementation.
Signature	<code>class SystemClock(Clock): def now(self) -> datetime</code>
Post	<code>now()</code> returns <code>datetime.now(timezone.utc)</code> .

FixedClock

Purpose	Returns a single fixed time. Used in deterministic unit tests.
Signature	<code>class FixedClock(Clock): __init__(fixed: datetime) ; set(t: datetime) -> None</code>
Pre	<code>fixed</code> must be tz-aware.

SimulatedClock

Purpose	Monotonic clock for backtests.
Signature	<code>class SimulatedClock(Clock): __init__(start: datetime) ; advance_to(t: datetime) -> None</code>
Pre	<code>advance_to(t)</code> requires $t \geq$ current internal time.
Post	Internal time is set to <code>t</code> .
Raises	<code>ValueError</code> if <code>t</code> would rewind time.

2.2 core.enums

Side (BUY, SELL)

Purpose	Order side; carries arithmetic sign for position math.
Signature	<code>side.sign -> int</code> # +1 for BUY, -1 for SELL
Signature	<code>Side.from_delta(delta: int) -> Side</code>
Raises	<code>ValueError</code> when <code>delta == 0</code> .

OrderType

Members	MKT, LMT, STP, STP_LMT.
----------------	-------------------------

TimeInForce

Members	DAY, GTC, IOC, FOK, GTD.
----------------	--------------------------

OrderStatus

Purpose	Venue-neutral lifecycle status, mapped from EMSX route status.
Members	NEW, RISK_APPROVED, RISK_REJECTED, SENT, ACCEPTED, WORKING, PART_FILLED, FILLED, CANCEL_PENDING, CANCELLED, MODIFY_PENDING, REJECTED, ROUTE_ERR, UNKNOWN.
Const	TERMINAL_ORDER_STATUSES = \{FILLED, CANCELLED, REJECTED, ROUTE_ERR\}.

RouteStatus, KillSwitchState

RouteStatus	Mirror of EMSX route status strings: SENT, WORKING, PARTFILLED, FILLED, CANCEL, REJECTED, ROUTE-ERR.
KillSwitchState	ARMED (trading allowed) or TRIPPED (all orders rejected).

2.3 core.exceptions

All custom exceptions inherit **FESError**. Layer-specific subclasses:

ConfigError	invalid or missing config.
MarketDataError	market-data session/subscription failure.
StaleMarketDataError	subclass for staleness specifically.
StrategyError	strategy invariant violation.
OrderError	bad order intent.
DuplicateOrderError	re-submission of a known idempotency key.
RiskRejection	raised when a hard check fails; <code>.reasons: list[str]</code> .
ExecutionError	generic venue error.
EMSXError	EMSX-specific.
ReconciliationBreak	internal vs. external positions disagree.

2.4 core.identifiers

new_uuid, correlation_id

Signature	<code>new_uuid() -> str</code>
Signature	<code>correlation_id(prefix: str = "corr") -> str</code>
Post	Returns globally unique strings.

idempotency_key

Purpose	Deterministic key used by OrderManager to dedupe submissions.
Signature	<code>idempotency_key(strategy_id: str, instrument: str, bar_end: datetime, side: str, qty: int) -> str</code>
Post	Same inputs $\Rightarrow$ same key (SHA-1, truncated). Replays produce identical keys.

emsx_request_id

Signature	<code>emsx_request_id(order_id: str, action: str) -> str</code>
Post	Unique correlation tag for an EMSX request.

2.5 core.logging

configure_logging, get_logger

Signature	<code>configure_logging(level: str = "INFO", json: bool = True) -> None</code>
Signature	<code>get_logger(name: str) -> structlog.BoundLogger</code>
Post	All logs are structured (key/value), ISO-8601 UTC timestamped, level-filtered.

2.6 core.events

All events are frozen @dataclass(slots=True, frozen=True); immutable across the bus.

MarketTick

Fields	<code>instrument, bid?, ask?, last?, volume?, exchange_timestamp?, receive_timestamp.</code>
Note	? = optional. <code>receive_timestamp</code> is always set by the provider.

BarClosed

Fields	<code>instrument, start_time, end_time, open, high, low, close, volume, interval_minutes=1.</code>
Invariant	<code>low <= open, close <= high; end_time > start_time.</code>

TargetPosition

Fields	<code>strategy_id, instrument, target_qty: int, timestamp, reason, metadata.</code>
Invariant	<code>target_qty</code> is signed; positive means net long.

OrderIntent

Fields	<code>strategy_id, instrument, side, qty (>0), order_type, time_in_force, idempotency_key, source_timestamp, limit_price?, stop_price?, metadata.</code>
Invariant	<code>qty > 0</code> . Direction lives in <code>side</code> , not the sign of <code>qty</code> .

RiskDecision

Fields	<code>order_key, approved: bool, reasons: tuple[str,...], decided_at.</code>
Factories	<code>RiskDecision.approve(order_key, decided_at), RiskDecision.reject(order_key, reasons, decided_at).</code>

ExecutionAck, ExecutionUpdate, FillUpdate, KillSwitchEvent

ExecutionAck	Synchronous response of submit/cancel/modify: order_id, route_id?, venue_request_id?, accepted: bool, message, timestamp.
ExecutionUpdate	Async push from venue subscription: order_id, route_id?, instrument, status (OrderStatus), filled_qty, avg_price?, leaves_qty, timestamp, raw.
FillUpdate	A trade execution: order_id, route_id?, instrument, side, fill_qty, fill_price, timestamp.
KillSwitchEvent	tripped, reason, timestamp, actor.

3 Module config/

Strongly typed pydantic models loaded from six YAML files under config/.

load_yaml, load_app_config

Signature	load_yaml(path: str Path) -> dict[str, Any]
Signature	load_app_config(config_dir: str Path) -> AppConfig
Pre	config_dir contains all six YAMLs (instruments, strategies, risk_limits, emsx, market_data, environments).
Post	Returns validated AppConfig ; raises ConfigError on missing/invalid files.

InstrumentConfig

Fields	symbol, bloomberg_topic, exchange, currency, tick_size, point_value, min_qty, max_qty, roll_days_before_expiry, session_tz, session_open, session_close.
Invariant	tick_size, point_value > 0; 1 <= min_qty <= max_qty.

InstrumentsConfig

Signature	by_symbol(symbol: str) -> InstrumentConfig
Raises	ConfigError if symbol not configured.

StrategyConfig, StrategiesConfig

StrategyConfig	strategy_id, type, instrument, interval_minutes=1, base_qty=1, params: dict.
StrategiesConfig	holds strategies: list[StrategyConfig].

RiskLimitsConfig

Fields	max_order_qty, max_position, max_notional, max_orders_per_minute=10, max_cancels_per_minute=20, stale_data_seconds=30, require_market_session=true, kill_switch_armed=true.
---------------	---

EMSXConfig

Fields	host="localhost", port=8194, service="//blp/emapisvc_beta", use_server_api=True, trader_uuid?, account?, broker="BMTB", handling_instruction="ANY", strategy="VWAP", default_order_type="MKT", default_tif="DAY".
Invariant	service must start with //blp/.

MarketDataConfig, EnvironmentsConfig, AppConfig

MarketDataConfig	provider, host, port, service, fields, historical_service, bar_interval_minutes.
EnvironmentsConfig	name, paper_trading, db_url, event_log_path, bus, bus_url?, metrics_port.
AppConfig	aggregate of all six configs.

4 Module market_data/

4.1 market_data.base

MarketDataProvider (abstract)

Purpose The single seam that decouples the strategy from Bloomberg. All implementations conform to this interface.

Signature

```
class MarketDataProvider(ABC):
    def start(self) -> None
    def stop(self) -> None
    def subscribe(self, instruments: list[str], fields: list[str]) -> None
    def on_tick(self, callback: Callable[[MarketTick], None]) -> None
    def request_historicalBars(self, instrument, start, end, interval_minutes=1) ->
        Iterable[BarClosed]
```

Contract

- `start` initializes the session. Must be called before `subscribe`.
- `subscribe` is idempotent: re-subscribing the same instrument has no effect.
- `on_tick` callbacks are invoked once per `MarketTick` in arrival order; exceptions in callbacks must not kill the provider thread.
- `stop` releases the session; further calls are no-ops.

4.2 market_data.bloomberg_provider

BloombergMarketDataProvider

Purpose BLPAPI implementation of `MarketDataProvider`. Threaded event loop pulls events from the session and dispatches `SUBSCRIPTION_DATA` events to callbacks.

Signature

```
__init__(host="localhost", port=8194,
         service="//blp/mktdata",
         historical_service="//blp/refdata")
```

Pre `blpapi` is installed; reachable Bloomberg session (Desktop API, SAPI, or B-PIPE).

Raises `MarketDataError` if session start or service open fails, or if `blpapi` is missing.

Fields supported

BID, ASK, LAST_PRICE, VOLUME, EVT_TIME_STAMP (any subset).

4.3 market_data.mock_provider

MockMarketDataProvider

Purpose Test and paper-trading replacement; ticks are pushed manually.

Signature `push_tick(tick: MarketTick) -> None`

Signature `MockMarketDataProvider.tick(instrument, last, ts=None, volume=1) -> MarketTick`

Post Synchronously delivers the tick to every registered callback.

4.4 market_data.bar_builder

MinuteBarBuilder

Purpose	Streaming OHLCV aggregator. Closes the current bar deterministically when a tick crosses into the next minute (or <code>flush()</code> is called).
Signature	<pre>__init__(interval_minutes: int = 1) on_tick(tick: MarketTick) -> None on_bar(callback: Callable[[BarClosed], None]) -> None flush(instrument: str None = None) -> None</pre>
Contract	• <code>on_tick</code> uses <code>exchange_timestamp</code> when present, else <code>receive_timestamp</code>. • Price source order: <code>last</code> → mid of (<code>bid</code>, <code>ask</code>); tick is dropped if neither is available. • Bars emitted only when a strictly later minute arrives. Use <code>flush</code> at shutdown to emit pending bars.
Raises	<code>ValueError</code> if <code>interval_minutes <= 0</code> .

4.5 market_data.tick_store

InMemoryTickStore

Purpose	Bounded per-instrument ring buffer of recent ticks (default 1024).
Signature	<pre>append(tick: MarketTick) -> None last(instrument: str) -> MarketTick None recent(instrument: str, n: int = 100) -> list[MarketTick] instruments() -> list[str]</pre>
Concurrency	All operations are thread-safe.

4.6 market_data.contract_resolver

ContractResolver

Purpose	Resolves logical symbols to active contracts and decides when to roll.
Signature	<pre>resolve(symbol: str, as_of: date None = None) -> InstrumentConfig should_roll(ins: InstrumentConfig, as_of: date, expiry: date) -> bool known_symbols() -> list[str] validate(symbol: str) -> None # raises ConfigError if unknown</pre>

4.7 market_data.stale_data_monitor

StaleDataMonitor

Purpose	Tracks the freshness of incoming ticks; consumed by <code>PreTradeRiskGateway</code> .
Signature	<pre>__init__(max_age_seconds: int, clock: Clock None = None) on_tick(tick: MarketTick) -> None last_seen(instrument: str) -> datetime None is_stale(instrument: str) -> bool age_seconds(instrument: str) -> float None</pre>
Contract	• An instrument that has never been seen is stale. • Naive timestamps are treated as UTC for the age computation.

5 Module strategy/

5.1 strategy.base

PortfolioView (Protocol)

Purpose Read-only window the strategy is allowed to see.

Signature

```
class PortfolioView(Protocol):
    def position(self, instrument: str) -> int
    def avg_cost(self, instrument: str) -> float | None
```

Strategy (abstract)

Purpose Consumes bars, emits target positions. Pure logic; no I/O.

Signature

```
class Strategy(ABC):
    strategy_id: str
    def on_bar(self, bar: BarClosed, portfolio: PortfolioView) -> list[TargetPosition]
    def warmup(self, bars: list[BarClosed]) -> None # optional
```

Contract

- `on_bar` must be pure with respect to external state apart from the strategy's own indicator state.
- Targets returned must reference `bar.instrument` or any instrument the strategy declared at init.

5.2 strategy.indicators

SMA, EMA, RollingZ

SMA `__init__(period: int); update(x: float) -> float | None; ready: bool.` Returns None until period observations have been seen.

EMA Same shape; smoothing factor $\alpha = 2/(period + 1)$.

RollingZ `__init__(period >= 2);` returns 0.0 when sample stdev is zero, else $(x - \mu)/\sigma$.

Raises `ValueError` for non-positive periods.

5.3 strategy.signal_state

SignalState

Purpose Bundles fast/slow EMA indicators into one object.

Signature

```
SignalState(fast_lookback: int, slow_lookback: int)
update(close: float) -> float | None # fast - slow, None until both ready
ready: bool
```

5.4 strategy.target_position

build_target_position

Purpose	Pure function: signal $\rightarrow$ <code>TargetPosition</code> .
Signature	<pre>build_target_position(strategy_id: str, instrument: str, signal: float, timestamp: datetime, base_qty: int, entry_threshold: float = 0.0, exit_threshold: float = 0.0, max_position: int None = None,) -> TargetPosition</pre>
Behaviour	signal > entry_threshold $\Rightarrow$ +base_qty; signal < -entry_threshold $\Rightarrow$ -base_qty; otherwise 0. Clipped to [-max_position, +max_position] if given.

5.5 strategy.minute_strategy

MinuteFuturesStrategy

Purpose	Reference EMA-crossover strategy on minute bars.
Signature	<pre>MinuteFuturesStrategy(strategy_id: str, instrument: str, base_qty: int, params: dict None = None)</pre>
Params keys	fast_lookback (default 5), slow_lookback (default 20), entry_threshold, exit_threshold, max_position_contracts.
Post	Each on_bar call returns either [] (warmup) or one <code>TargetPosition</code> for self.instrument.

5.6 strategy.roll_logic

RollLogic

Signature	<pre>RollLogic(instrument: InstrumentConfig) evaluate(active_contract: str, next_contract: str, active_expiry: date, as_of: date) -> RollDecision</pre>
RollDecision	should_roll, from_contract, to_contract?, reason.
Rule	Roll when (expiry - as_of).days <= roll_days_before_expiry.

6 Module portfolio/

6.1 portfolio.positions

PositionBook

Purpose Authoritative in-memory net position + weighted-average cost per instrument. Updated by fills.

Signature

```
position(instrument: str) -> int
avg_cost(instrument: str) -> float | None
apply_fill(fill: FillUpdate) -> None
snapshot() -> dict[str, tuple[int, float]] # symbol -> (qty, avg_cost)
set(instrument: str, qty: int, avg_cost: float = 0.0) -> None
is_flat(instrument: str) -> bool
```

- Avg-cost rule**
- Adding to an existing position $\Rightarrow$ weighted average.
 - Reducing $\Rightarrow$ avg unchanged unless flipped.
 - Flipping side $\Rightarrow$ new avg = fill price.

Concurrency Thread-safe (internal lock).

6.2 portfolio.fills

FillLedger

Purpose Append-only fill record, indexed by order_id.

Signature

```
record(fill: FillUpdate) -> None
for_order(order_id: str) -> list[FillUpdate]
all() -> list[FillUpdate]
total_filled(order_id: str) -> int
```

6.3 portfolio.pnl

PnLCalculator

Purpose Realized PnL on closing trades + unrealized PnL against a mark map. Uses each instrument's point_value.

Signature

```
__init__(instruments: InstrumentsConfig)
apply_fill(fill: FillUpdate) -> None
realized() -> float
unrealized(marks: dict[str, float]) -> float
snapshot(marks: dict[str, float]) -> PnLSnapshot # (realized, unrealized, total)
```

Math On closing n contracts of a long at avg $\bar{p}$ at price p : realized $+= n \cdot (p - \bar{p}) \cdot \text{pointValue}$.

6.4 portfolio.exposure

ExposureCalculator

Signature

```
notional(instrument: str, qty: int, price: float) -> float
gross_notional(positions: PositionBook, marks: dict[str, float]) -> float
per_instrument(positions, marks) -> list[Exposure]
```

Formula

$\text{notional} = |qty| \cdot \text{price} \cdot \text{pointValue}.$

6.5 portfolio.reconciliation

Reconciler**Signature**

```
compare(external_positions: dict[str, int]) -> list[PositionBreak]
force_resync(external_positions: dict[str, int]) -> None
```

PositionBreak instrument, internal_qty, external_qty; computed delta = external - internal.

Note force_resync is destructive and trusts the external source — requires explicit operator approval.

7 Module orders/

7.1 orders.models

WorkingOrderBook

Purpose	Net signed working quantity per instrument across all open orders. Consumed by OrderManager to avoid double-sending.
Signature	<pre> upsert(order_id, instrument, side, leaves_qty, status) -> None remove(order_id: str) -> None net_working_qty(instrument: str) -> int # signed snapshot() -> list[tuple[order_id, sym, Side, qty, OrderStatus]] </pre>
Auto-removal	Terminal statuses (FILLED, CANCELLED, REJECTED, ROUTE_ERR) or leaves_qty<=0 remove the row.

7.2 orders.idempotency

IdempotencyBackend (protocol), IdempotencyStore

IdempotencyStore

```

seen(key: str) -> bool
mark(key: str) -> None
claim(key: str) -> bool    # True iff key was new (and is now marked)

```

Default backend

In-memory; swap in a Redis/SQL backend by passing an alternate **IdempotencyBackend**.

7.3 orders.throttles

RateLimiter

Purpose Sliding-window event limiter (e.g. orders/minute).

Signature

```

__init__(max_events: int, window_seconds: int, clock: Clock | None = None)
try_acquire() -> bool    # True if event accepted
current() -> int         # events currently in window

```

7.4 orders.sizing

slice_order

Purpose Break a large parent qty into child clips.

Signature slice_order(total_qty: int, max_clip: int) -> list[int]

Post sum(result) == total_qty; all(c <= max_clip); returns [] when total_qty <= 0.

Raises ValueError when max_clip <= 0.

7.5 orders.lifecycle

OrderRecord, OrderLifecycle

OrderRecord order_id, strategy_id, instrument, side, qty, idempotency_key, created_at, status, venue_order_id?, route_id?, filled_qty=0, avg_price?, last_update?, reasons[].

OrderLifecycle

```

register(rec: OrderRecord) -> None
get(order_id: str) -> OrderRecord | None
update_status(order_id, status, filled_qty?, avg_price?,
               venue_order_id?, route_id?, timestamp?) -> OrderRecord | None
all() -> list[OrderRecord]
by_venue_id(venue_order_id: str) -> OrderRecord | None

```

7.6 orders.order_manager

OrderManager

Purpose Pure deduplicating mapping `TargetPosition` $\rightarrow$ list of `OrderIntent`.

Signature

```

__init__(positions: PositionBook, working: WorkingOrderBook,
          idempotency: IdempotencyStore,
          default_order_type: OrderType = OrderType.MKT,
          default_tif: TimeInForce = TimeInForce.DAY,
          max_clip: int | None = None)

on_target(target: TargetPosition) -> list[OrderIntent]

```

Algorithm

1. $\Delta = \text{target.target_qty} - \text{position} - \text{net working qty}$.
2. If $\Delta = 0$, return [].
3. Otherwise `side = Side.from_delta( $\Delta$ ), qty =  $|\Delta|$` .
4. Optionally slice with `max_clip`.
5. For each clip, compute a deterministic `idempotency_key`; submit only if `idempotency.claim(key)` returns True.

8 Module risk/

8.1 risk.limits

RiskLimits

Purpose	Frozen dataclass mirror of RiskLimitsConfig used by the gateway.
Fields	max_order_qty, max_position, max_notional, stale_data_seconds, require_market_session, max_orders_per_minute, max_cancels_per_minute.

8.2 risk.validations

ValidationCheck

Purpose	Reusable pass/fail check result.
Signature	ValidationCheck.ok(name) / ValidationCheck.fail(name, reason).
Fields	name, passed: bool, reason: str.

8.3 risk.session_checks

SessionChecker

Signature	is_open(now_utc: datetime) -> bool
Contract	Handles wrap-around sessions where close < open (overnight). Uses the instrument's session_tz .

8.4 risk.kill_switch

KillSwitch

Signature	<pre> __init__(armed: bool = True, clock: Clock None = None) is_tripped -> bool # property trip(reason: str, actor: str = "system") -> KillSwitchEvent arm(actor: str = "operator") -> KillSwitchEvent reason -> str None # property </pre>
Contract	When tripped, the pre-trade gateway rejects every order. Tripping is logged at error level; arming at warning .

8.5 risk.pre_trade

PreTradeRiskGateway

Purpose Mandatory gate. Every **OrderIntent** must pass before reaching **ExecutionAdapter**.

Signature

```
__init__(
    limits: RiskLimits,
    instruments: InstrumentsConfig,
    positions: PositionBook,
    exposure: ExposureCalculator,
    stale_monitor: StaleDataMonitor,
    kill_switch: KillSwitch,
    order_rate: RateLimiter,
    get_mark: Callable[[str], float | None] | None = None,
    clock: Clock | None = None,
)

validate(intent: OrderIntent) -> CheckResult
```

Checks (all must pass) • **kill_switch** — rejects if tripped.

- **max_order_qty** — $1 \leq \text{qty} \leq \text{limit}$.
- **max_position** — projected $|\text{pos}|$ after fill $\leq \text{limit}$.
- **max_notional** — $|\text{qty}| \cdot \text{mark} \cdot \text{pointValue} \leq \text{limit}$. Skipped when mark is unknown.
- **market_session** — skipped if **require_market_session** is false.
- **stale_market_data** — rejects if last tick older than **stale_data_seconds**.
- **order_rate_limit** — consumes a slot from the **order_rate** limiter.

CheckResult **decision**: RiskDecision, **checks**: tuple[ValidationCheck,...].

9 Module execution/

9.1 execution.base

ExecutionAdapter (abstract)

Purpose Single interface for any venue (EMSX, FIX, paper).

Signature

```
class ExecutionAdapter(ABC):
    def start(self) -> None
    def stop(self) -> None
    def submit_order(self, order: OrderIntent) -> ExecutionAck
    def cancel_order(self, order_id: str) -> ExecutionAck
    def modify_order(self, order_id: str, changes: dict) -> ExecutionAck
    def on_execution_update(self, callback: ExecutionUpdateCallback) -> None
    def on_fill(self, callback: FillCallback) -> None
```

Contract

- Sync calls (submit/cancel/modify) return an **ExecutionAck** immediately; persistent state is delivered asynchronously via callbacks.
- Multiple callbacks may be registered for the same event type; all are invoked.

9.2 execution.emsx_mapper

EMSXMapper

Purpose Only place that knows EMSX wire field names. Translates both directions.

Signature

```
to_create_order_and_route(intent: OrderIntent) -> dict[str, Any]
to_modify_route(route_id: str, changes: dict) -> dict[str, Any]
to_cancel_route(route_id: str) -> dict[str, Any]
from_create_response(response: dict, intent: OrderIntent) -> ExecutionAck
from_route_subscription(msg: dict) -> ExecutionUpdate
from_fill_message(msg: dict) -> FillUpdate
```

Status mapping

```
{SENT      → SENT,WORKING      → WORKING,PARTFILLED      →
PART_FILLED,FILLED → FILLED,CANCEL → CANCELLED,REJECTED →
REJECTED,ROUTE-ERR → ROUTE_ERR}.
```

9.3 execution.emsx_requests

EMSXRequests

Purpose Thin BLPAPI wrapper for the EMSX request/response service.

Signature

```
start() / stop()
create_order_and_route(fields: dict) -> dict      # CreateOrderAndRouteEx
create_order(fields: dict) -> dict                # CreateOrder
route_ex(fields: dict) -> dict                    # RouteEx
modify_route(fields: dict) -> dict                # ModifyRouteEx
cancel_route(fields: dict) -> dict                # CancelRouteEx
```

Raises **EMSXError** on session/service open failure or when **blpapi** is missing.

Note Unknown fields are logged but do not abort the request — EMSX schema evolution must not break callers.

9.4 execution.emsx_subscriptions

EMSXSubscriptions

Purpose	Subscribes to <code>//blp/emapisvc/order</code> and <code>//blp/emapisvc/route</code> ; dispatches each message to typed callbacks.
Signature	<pre>__init__(host, port, order_topic, route_topic, order_fields: list[str] None, route_fields: list[str] None) start() / stop() on_order_update(cb: Callable[[dict], None]) -> None on_route_update(cb: Callable[[dict], None]) -> None</pre>
Defaults	Order/Route fields cover <code>EMSX_SEQUENCE</code> , <code>EMSX_ROUTE_ID</code> , <code>EMSX_TICKER</code> , <code>EMSX_SIDE</code> , <code>EMSX_AMOUNT</code> , <code>EMSX_STATUS</code> , <code>EMSX_FILLED</code> , <code>EMSX_WORKING</code> , <code>EMSX_AVG_PRICE</code> , <code>EMSX_FILL_AMOUNT</code> , <code>EMSX_FILL_PRICE</code> ,
Resilience	Callback exceptions are caught and logged; the event loop survives.

9.5 execution.emsx_state_machine

EMSXStateMachine

Purpose	Validates that incoming venue-status transitions are legal. Detects out-of-order subscription messages.
Signature	<pre>current(order_id: str) -> OrderStatus transition(order_id: str, new: OrderStatus) -> bool # False if illegal on_violation(cb: Callable[[order_id, current, attempted], None]) -> None</pre>
Rules	Transitions out of terminal states are violations. Allowed transitions encoded in <code>_ALLOWED</code> table.

9.6 execution.emsx_adapter

EMSXExecutionAdapter

Purpose	High-level adapter; composes <code>EMSXRequests</code> , <code>EMSXSubscriptions</code> , <code>EMSXMapper</code> , <code>EMSXStateMachine</code> .
Signature	<pre>__init__(config: EMSXConfig, auto_route: bool = True, requests: EMSXRequests None = None, subscriptions: EMSXSubscriptions None = None) start() / stop() submit_order(order: OrderIntent) -> ExecutionAck cancel_order(order_id: str) -> ExecutionAck modify_order(order_id: str, changes: dict) -> ExecutionAck on_execution_update(cb) / on_fill(cb)</pre>
Routing modes	<code>auto_route=True</code> → <code>CreateOrderAndRouteEx</code> (Phase 3+). <code>auto_route=False</code> → <code>CreateOrder</code> only; trader routes manually in EMSX (Phase 2 staging).

9.7 execution.paper_adapter

PaperExecutionAdapter

Purpose	Internal paper-fill engine. No external wire.
Signature	<pre>__init__(get_mark: Callable[[str], float None], slippage_ticks: float = 0.0, tick_size_lookup: Callable[[str], float] None = None)</pre>
Behaviour	• On submit: emits SENT update, then a FillUpdate, then FILLED update. • Fill price: <code>limit_price</code> for LMT; <code>mark + slippage_ticks * tick_size * side.sign</code> otherwise. • Modifies are not supported.

10 Module storage/

10.1 storage.db

Database, get_database

Purpose	Minimal SQLite wrapper with shared lock and WAL journal. Schema initialized from <code>schema.sql</code> on construction.
Signature	<pre> Database(url: str) # only sqlite:/// supported execute(sql, params=()) -> None executemany(sql, params_list) -> None query(sql, params=()) -> list[sqlite3.Row] cursor() -> context manager close() -> None get_database(url: str None = None) -> Database # singleton helper </pre>
Raises	<code>ValueError</code> on non-sqlite URLs.

10.2 storage.event_log

EventLog (abstract), JsonlEventLog, SqliteEventLog

Purpose	Append-only audit trail. Strategy/risk/execution decisions are persisted.
Signature	<code>append(event_type: str, event: Any, correlation_id: str None = None) -> None</code>
Encoding	Dataclasses are converted via <code>asdict</code> ; Enums use <code>.value</code> ; datetimes use ISO-8601.
Thread-safety	<code>JsonlEventLog</code> uses an internal lock; safe to share across threads.

10.3 storage.repositories

PositionRepository, OrderRepository, FillRepository, RiskDecisionRepository

PositionRepository	<code>upsert(instrument, qty, avg_cost), get, all()</code> .
OrderRepository	<code>upsert(rec: OrderRecord, leaves_qty), by_idempotency_key(key)</code> for crash-restart dedupe.
FillRepository	<code>insert(fill), for_order(order_id)</code> .
RiskDecisionRepository	<code>insert(decision: RiskDecision)</code> .
idempotency_key uniqueness	Enforced by a <code>UNIQUE</code> constraint in <code>working_orders</code> .

10.4 storage.snapshots

SnapshotStore

Signature	<pre> save(name: str, payload: dict) -> None latest(name: str) -> dict None </pre>
Use	Periodic blob snapshots of risk state, strategy state, etc., for warm restarts.

11 Module messaging/

11.1 messaging.bus

EventBus (abstract)

Purpose Pub/sub interface used by every service.

Signature

```
class EventBus(ABC):
    def publish(self, topic: str, payload: Any) -> None
    def subscribe(self, topic: str, handler: Handler) -> None
    def start(self) -> None
    def stop(self) -> None
```

Handler Callable[[topic: str, payload: Any], None].

11.2 messaging.in_memory_bus

InMemoryBus

Semantics Synchronous. `publish` invokes all subscribed handlers inline.

Use Single-process deployments and tests.

Failures Handler exceptions are logged; never propagate to the publisher.

11.3 messaging.kafka_bus

KafkaBus

Signature `KafkaBus(bootstrap: str, group_id: str = "fes")`

Encoding Payloads JSON-serialized (dataclasses via `asdict`). Background consumer thread invokes handlers.

Use Multi-service deployments.

Optional dep `confluent-kafka`.

11.4 messaging.redis_bus

RedisStreamBus

Signature `RedisStreamBus(url: str, group: str = "fes", consumer: str = "fes-1")`

Semantics Consumer-group reads via `XREADGROUP`; `XACK` after handler success.

Optional dep `redis`.

make_bus (factory)

Signature `make_bus(kind: str, url: str | None = None) -> EventBus`

kind "memory", "kafka" (requires url), "redis" (requires url).

Raises `ValueError` for unknown kind or missing URL.

12 Module monitoring/

12.1 monitoring.metrics

Metrics

Purpose	Prometheus counters/gauges/histograms exposed on the HTTP endpoint.
Signature	<code>Metrics(port: int None = 9100)</code> (pass <code>None</code> to disable the HTTP server)
Counters	<code>ticks_total</code> , <code>bars_total</code> , <code>signals_total</code> , <code>orders_submitted</code> , <code>orders_rejected</code> , <code>fills_total</code> .
Gauges	<code>open_position</code> , <code>working_qty</code> , <code>realized_pnl</code> , <code>unrealized_pnl</code> , <code>kill_switch_tripped</code> , <code>market_data_age_seconds</code> .
Histogram	<code>bar_to_order_seconds</code> (buckets 5ms ... 5s).

12.2 monitoring.health

HealthProbe, HealthChecker

HealthProbe	<code>name</code> , <code>check: Callable[[], bool]</code> , <code>detail: Callable[[], str]</code> .
HealthChecker	<pre> register(probe: HealthProbe) -> None check_all() -> dict[name -> {ok: bool, detail: str}] is_healthy() -> bool </pre>

12.3 monitoring.alerts

AlertSink, ConsoleAlertSink

Signature	<code>alert(severity: str, message: str, **fields) -> None</code>
Severity	<code>critical</code> , <code>error</code> , <code>warning</code> , <code>info</code> .
Implementations	<code>ConsoleAlertSink</code> (default). <code>Slack/PagerDuty</code> subclasses pluggable.

13 Module backtest/

13.1 backtest.slippage

SlippageModel (abstract), FixedTickSlippage

Signature `adjust(mid: float, side: Side, instrument: str) -> float`

FixedTickSlippage

`__init__(ticks: float, tick_size_lookup); mid + ticks*tick_size*side.sign.`

13.2 backtest.commission

CommissionModel (abstract), FlatCommissionModel

Signature `per_fill(qty: int, price: float, instrument: str) -> float`

FlatCommissionModel

`__init__(per_contract: float = 1.25); |qty| * per_contract.`

13.3 backtest.replay

BarReplay, TickReplay

BarReplay `__iter__()` sorted by `end_time`; `between(start, end)` yields a subrange.

TickReplay `__iter__()` sorted by `exchange_timestamp` or `receive_timestamp`.

13.4 backtest.engine

BacktestEngine

Purpose Drives the same **Strategy** + **OrderManager** + **PositionBook** components used live. Fills are simulated.

Signature

```
__init__(strategy: Strategy, instruments: InstrumentsConfig,
         slippage: SlippageModel | None = None,
         commission: CommissionModel | None = None)

run(bars: Iterable[BarClosed]) -> BacktestResult
```

BacktestResult `bars_processed, targets_emitted, intents_emitted, fills_count, realized_pnl, unrealized_pnl, total_pnl, total_commission, end_positions: dict[str,int], timeline: list[(datetime, PnLSnapshot)]`.

Guarantee Live $\equiv$ backtest at the strategy + risk + order layer (same code paths).

13.5 backtest.reports

generate_report

Signature `generate_report(result: BacktestResult) -> dict[str, Any]`

Keys returned

`bars, targets, intents, fills, realized_pnl, unrealized_pnl, total_pnl,`
`total_commission,`
`net_pnl, sharpe_annualized, max_drawdown, end_positions.`

Sharpe convention

Per-bar returns scaled by $\sqrt{252 \cdot 390}$ minutes/year.

14 Module app/

14.1 app.topics

Topic name constants

Constants	TICKS, BARS, TARGETS, INTENTS, RISK_DECISIONS, EXECUTION_UPDATES, FILLS, KILL_SWITCH.
Rule	All services use <i>only</i> these constants when publishing/subscribing.

14.2 app.wiring

Services, build_services

Purpose	Single composition root. Every entry point builds the same DI graph.
Signature	

```
build_services(
    config_dir: str | Path, *,
    metrics_port: int | None = None,
    market_data: MarketDataProvider | None = None,
    execution: ExecutionAdapter | None = None,
) -> Services
```

Services fields	config, bus, db, event_log, metrics, health, alerts, market_data, bar_builder, tick_store, stale_monitor, positions, fills, pnl, exposure, working, idempotency, order_manager, kill_switch, risk, execution, strategies.
Defaults	MockMarketDataProvider and PaperExecutionAdapter when not overridden — dev/test/paper trading work out of the box.

14.3 app.run_market_data

Entry point: run_market_data

CLI	python -m futures_emsx_strategy.app.run_market_data --config-dir config [--bloomberg --mock] [--metrics-port N].
Behaviour	- Subscribes to instruments listed in <code>instruments.yaml</code>. - For each tick: updates <code>InMemoryTickStore</code>, <code>StaleDataMonitor</code>, <code>MinuteBarBuilder</code>, then publishes <code>TICKS</code>; <code>BARS</code> on close. - Persists <code>MarketTick</code> and <code>BarClosed</code> to the event log.

14.4 app.run_strategy

Entry point: run_strategy

CLI	python -m futures_emsx_strategy.app.run_strategy --config-dir config.
Behaviour	Subscribes to <code>BARS</code> ; routes each bar to all strategies bound to that instrument; publishes <code>TARGETS</code> for each <code>TargetPosition</code> emitted.

14.5 app.run_execution

Entry point: run_execution

CLI ... --config-dir config [--emsx|--paper] [--auto-route|--stage-only] [--metrics-port N].

Pipeline (per TARGETS message)

1. **OrderManager.on_target** → intents.
2. Publish **INTENTS**; persist; run **PreTradeRiskGateway.validate**; persist + publish **RISK_DECISIONS**.
3. If approved, call **execution.submit_order**; update **WorkingOrderBook** and **OrderLifecycle**.
4. Inbound **ExecutionUpdate/FillUpdate** update lifecycle, working book, positions, PnL, metrics; publish **EXECUTION_UPDATES**, **FILLS**.

14.6 app.run_reconciler

Entry point: run_reconciler

CLI ... --config-dir config --external-positions broker.json --interval-seconds 60.

Behaviour Periodically loads `\{symbol: qty\}` from the file and compares against **PositionBook**. Emits warning alerts on breaks. Never auto-resyncs.

14.7 app.run_all_local

Entry point: run_all_local

CLI ... --config-dir config [--emit-ticks|--no-emit-ticks] [--seconds N].

Behaviour One-process composition over **InMemoryBus**. Used for paper trading, dev, integration testing. With `--emit-ticks`, drives **MockMarketDataProvider** with a synthetic random walk.

15 Topology & guarantees

15.1 Service boundaries

Service	Subscribes to	Publishes
market-data	(Bloomberg, OS)	TICKS, BARS
strategy	BARS	TARGETS
execution	TARGETS	INTENTS, RISK_DECISIONS, EXECUTION_UPDATES, FILLS
reconciler	(filesystem, OS)	alerts
monitoring	(Prometheus scrape)	—

15.2 Idempotency & at-least-once delivery

The bus is treated as *at-least-once*. Idempotency is enforced inside the **OrderManager** via `idempotency\_key`:

```
key = SHA1(strategy_id | instrument | bar_end | side | qty)[:16]
```

If the same key has already been claimed by **IdempotencyStore**, the intent is suppressed. The same key is also stored in `working\_orders.idempotency\_key` (UNIQUE), giving database-level dedupe across restarts.

15.3 State-machine invariants

EMSXStateMachine only permits the transitions in `\_ALLOWED`. Notable rules:

- Terminal states (FILLED, CANCELLED, REJECTED, ROUTE_ERR) are sinks. Any further status change triggers a violation callback.
- NEW cannot jump directly to SENT; it must first be RISK_APPROVED.
- UNKNOWN is a recovery state — any transition back into known states is allowed, used after gap-fill on restart.

15.4 Phased rollout flags

Phase	Adapter	Notes
1	PaperExecutionAdapter	No external wire; full audit
2	EMSXExecutionAdapter <code>auto_route=False</code>	Stages orders; trader routes
3	EMSXExecutionAdapter <code>auto_route=True</code>	Tight <code>risk_limits.yaml</code> , one symbol, small size
4	Same as 3	Production limits, reconciler + kill switch live

15.5 Error budget for live trading

Any of the following must trip the kill switch (manually or via alert action):

- Market data age > `stale_data_seconds` (default 30) for any subscribed instrument.
- Sustained risk rejections (e.g., > 1/s for 5 min).
- State-machine violations (`EMSXStateMachine.on\_violation`).
- Position breaks reported by **Reconciler**.

16 Quick reference — minimum contracts to satisfy when extending

16.1 Adding a market-data vendor

Implement **MarketDataProvider** (see `market_data.base`). Bind it into `build_services` via the `market\_data` argument. No other layer changes.

16.2 Adding a venue

Implement **ExecutionAdapter**. Provide a venue-specific mapper analogous to **EMSXMapper**. Reuse **EMSXStateMachine** or write a venue-specific one. Bind via the `execution` argument to `build_services`.

16.3 Adding a strategy

Subclass **Strategy**. Read only via **PortfolioView**. Emit **TargetPosition**. Add an entry under `strategies.yaml`; the wiring layer instantiates it.

16.4 Adding a risk check

Add a private method `_check_*` to **PreTradeRiskGateway** that returns a **ValidationCheck**. Append it to the list inside `validate`. Tests must show that a failing case rejects the order with that check's name in `reasons`.

16.5 Adding a bus implementation

Implement **EventBus**. Register it in `make_bus`. The new transport must:

- Serialize dataclasses correctly (see **KafkaBus** pattern).
- Catch and log handler exceptions.
- Provide at-least-once delivery (so **IdempotencyStore** can dedupe).

17 Test contract surface

Every test in `futures\emsx\strategy/tests/` corresponds to a published invariant.

Test	Invariant being enforced
<code>unit/test_bar_builder.py</code>	Bars close only on a strictly later minute; <code>flush()</code> drains pending.
<code>unit/test_order_manager.py</code>	Δ math, working-qty offset, sign of <code>Side.from_delta</code> , idempotency dedupe.
<code>unit/test_pre_trade_risk.py</code>	Each risk check rejects exactly its scenario; happy path approves.
<code>unit/test_positions_pnl.py</code>	Weighted-average cost, realized PnL with point value, unrealized when flat.
<code>unit/test_state_machine.py</code>	Legal transition chain; post-terminal transitions are violations.
<code>unit/test_indicators.py</code>	Indicators return <code>None</code> until ready; constant series $\Rightarrow$ z-score 0.
<code>integration/test_end_to_end_paper.py</code>	Ticks $\rightarrow$ bars $\rightarrow$ strategy $\rightarrow$ risk $\rightarrow$ paper fills $\rightarrow$ position update.
<code>integration/test_backtest.py</code>	Backtest engine runs and reports keys exist.
<code>emsx_sim/test_emsx_adapter_fake.py</code>	Adapter pipeline against an in-process BLPAPI fake; route subscription $\rightarrow$ fill propagation.
<code>replay/test_replay_deterministic.py</code>	Same bar series $\Rightarrow$ identical orders + idempotency keys.

End of document.